

Utah.gov

WCAG 2.1 Accessibility Audit & EDS Migration Plan

CONFIDENTIAL REPORT

Target URL: <https://www.utah.gov>

Audit Date: March 12, 2026

Standard: WCAG 2.1 Level AA

Tool: axe-core 4.7.2 + Manual Testing

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1. Executive Summary

Overall Assessment

Utah.gov is **reasonably accessible** with 27 WCAG rules passing automated checks, but has **2 confirmed violations** and **2 areas requiring manual review**. The most impactful issues are the **missing H1 heading**, an **inaccessible logo link**, and **undersized touch targets**.

The site's card-heavy layout maps cleanly to Adobe Edge Delivery Services (EDS) block patterns, making it a strong candidate for migration. EDS enforces clean semantic HTML, proper heading hierarchy, and explicit content modeling — providing a natural opportunity to resolve all identified accessibility issues.

27

RULES
PASSED

2

VIOLATIONS

2

NEEDS
REVIEW

10

TOTAL ISSUES

2. WCAG 2.1 Accessibility Audit

Results

2.1 Critical & Serious Issues

#	Issue	WCAG Criterion	Severity	Details
1	Missing H1 heading	1.3.1 Info & Relationships (A)	Serious	The page has zero H1 elements. The heading hierarchy starts at H2/H3, breaking document outline for screen readers. 21 headings exist but none at level 1.
2	Logo link without accessible name	2.4.4 Link Purpose / 4.1.2 Name, Role, Value (A)	Serious	The main site logo link (<code>.utds-title-wrapper</code>) has no accessible text. No <code>aria-label</code> , no visible text, and the image inside is marked <code>role="presentation"</code> . Screen readers cannot announce this link's purpose.
3	16 images with empty alt text	1.1.1 Non-text Content (A)	Serious	Service icons (vehicle renewal, hunting, jobs, etc.) have <code>alt=""</code> but no <code>role="presentation"</code> . These are inside links with text context, but the pattern is inconsistent and some icons convey meaning.
4	Color contrast	1.4.3 Contrast Minimum (AA)	Serious	5 elements flagged for potential contrast

#	Issue	WCAG Criterion	Severity	Details
	issues (5 elements)			failures, including text overlaid on the hero background image and the search bar label. Requires manual verification against varying background images.
5	Focusable content inside aria-hidden	4.1.2 Name, Role, Value (A)	Serious	3 containers with <code>aria-hidden="true"</code> contain focusable elements: popup wrapper, vertical menu, and QSI feedback button. Keyboard users can tab into content hidden from screen readers.
6	5 unlabeled form inputs	1.3.1 / 4.1.2 (A)	Moderate	Google Translate widget injects 5 hidden text inputs with no labels or ARIA attributes (<code>goog-gt-votingInputSrcLang</code> , etc.).

2.2 Moderate Issues

#	Issue	WCAG Criterion	Severity	Details
7	Duplicate ID attribute	4.1.1 Parsing (A)	Minor	<code>id="home--localResources"</code> appears twice in the DOM, which can confuse assistive technology when referencing elements.
8	22 undersized touch targets	2.5.5 Target Size (AAA) / 2.5.8 Target Size Minimum (AA)	Moderate	22 of 102 interactive elements are below 44×44px. Examples: "Here is how you know" button (139×16px), "UtahID Sign In" (163×30px), persona pills (38px tall), hamburger menu (36×36px).

#	Issue	WCAG Criterion	Severity	Details
9	Auto-rotating hero with no pause	2.2.2 Pause, Stop, Hide (A)	Moderate	Background hero image cycles automatically with no visible pause/stop mechanism. Users with motion sensitivity or cognitive disabilities may be unable to consume content.
10	Invalid ARIA attribute value	4.1.2 Name, Role, Value (A)	Needs Review	1 element flagged by axe-core with a potentially invalid ARIA attribute value. Requires manual verification.

2.3 Things Done Well

- **PASS** Skip-to-main-content link is present and functional
- **PASS** Language attribute (`lang="en"`) is set on HTML element
- **PASS** Proper landmark regions: `<main>` , `<nav>` (2), `<header>` , `<footer>`
- **PASS** Viewport meta tag correctly configured (no zoom restriction)
- **PASS** External links in footer properly indicate "opens in a new tab"
- **PASS** Heading levels within sections follow proper H2 > H3 progression

- **PASS** Base font size is 16px (good for readability)
- **PASS** 27 automated accessibility rules passed

3. Page Structure Analysis for EDS Migration

3.1 Technology Stack

Property	Value
Framework	Utah Design System (UTDS) — Custom framework
React / Angular / Vue	None detected
CMS	Custom (not WordPress)
jQuery	Not loaded
Third-party widgets	Google Translate, Qualtrics Feedback, UtahID SSO
Responsive	Yes (width=device-width, initial-scale=1)

3.2 Section-to-Block Mapping

Original Section	Proposed EDS Block	Notes
Government banner ("Official website of the state of Utah")	Header (default content)	Standard government trust banner above header
Logo + UtahID Sign In + Hamburger menu	Header / Navigation	EDS header block with authentication CTA
Hero image + search bar (auto-rotating)	Hero block	Full-bleed image with overlaid search; rotating backgrounds need pause control
"I am:" persona chips	Cards (pill variant) or Tags block	Persona-based navigation with icon pills (6 options)
"Featured Services" grid (6 items)	Cards block	Icon + title + subtitle linked cards
"Utah Local" section (3 items)	Cards block (variant)	Same card pattern, fewer items
"Visiting Utah" section (3 items)	Cards block (variant)	Same card pattern with travel theme
Footer with logo + legal links	Footer	Standard footer with copyright and legal links

4. EDS Migration Plan with Accessibility Remediation

1 Foundation & Design System Setup

- Extract design tokens from UTDS (colors: navy #071D49, gold #F0B429, white; typography; spacing)
- Set up EDS project with `styles.css` and `scripts.js`
- Configure navigation (`nav.html`) from utah.gov header structure
- Define CSS custom properties with WCAG-compliant color pairs

2 Block Development (with Accessibility Fixes)

Block	Accessibility Fix Integrated
Header	Add <code>aria-label="Utah.gov Home"</code> to logo link; ensure hamburger menu has proper <code>aria-expanded</code> states; manage focus trapping in mobile menu
Hero	Add H1 heading; ensure search input has proper <code><label></code> ; add pause control for image rotation; enforce 4.5:1 contrast with semi-transparent overlay
Cards (services)	Add meaningful <code>alt</code> text to icons OR mark as <code>role="presentation"</code> ; enforce minimum 44x44px touch targets; ensure unique IDs
Cards (local/visit)	Same as above; proper heading hierarchy (H2 section > H3 cards)
Footer	Maintain "opens in new tab" announcements; ensure all links have accessible names

3 Content Migration

- Migrate homepage content into EDS document format
- Verify heading hierarchy: H1 (page title) > H2 (section titles) > H3 (card titles)
- Add proper image alt text for all service icons during content authoring
- Replace Google Translate widget with accessible alternative
- Test with screen reader (VoiceOver / NVDA) after migration

4 Accessibility-Specific Remediations

WCAG Issue	EDS Remediation Strategy
Missing H1	Add an H1 in the hero section. EDS default content supports this natively in the document model.
Logo link without name	In EDS header block JS, automatically add <code>aria-label</code> to the logo link during decoration.
Empty alt on icons	EDS content model makes alt text explicit per image during authoring. Authors set meaningful alt or mark decorative.
Color contrast	Define CSS custom properties with compliant color pairs. Add <code>background: linear-gradient(rgba(0,0,0,0.4), ...)</code> overlay on hero images.
Touch target size	Set <code>min-height: 44px; min-width: 44px</code> in block CSS for all interactive elements.
Focusable in aria-hidden	EDS block JS manages <code>aria-hidden</code> and <code>tabindex="-1"</code> together when toggling menus/popups.
Duplicate IDs	EDS content model prevents duplicate IDs by design (each block instance is unique).
Carousel pause	Implement <code>prefers-reduced-motion</code> media query and add visible pause button with <code>aria-label</code> .
Unlabeled inputs	Replace Google Translate widget with a lightweight accessible alternative with proper ARIA labeling.

5 Validation & Testing

- Run axe-core on all migrated pages — target **zero violations**
- Manual keyboard-only navigation test (Tab, Enter, Escape flows)
- Screen reader testing (NVDA + Chrome, VoiceOver + Safari)
- Lighthouse accessibility audit (target score: 95+)
- Verify touch targets at mobile viewport widths (320px, 375px, 414px)
- Test with `prefers-reduced-motion: reduce` enabled
- Validate color contrast across all hero image variations

5. Appendix: Detailed Element Audit

A. Images Audit (18 total, 16 issues)

Image Source	Issue
img/services/tax-rex.png	Empty alt, not marked presentation
img/services/dnr-wildlife.png	Empty alt, not marked presentation
img/services/dps-dld.png	Empty alt, not marked presentation
img/services/dws-jobs.png	Empty alt, not marked presentation
img/services/health-vitalrecords.png	Empty alt, not marked presentation
img/misc/def-more.png	Empty alt, not marked presentation
img/misc/gov-legislative.png	Empty alt, not marked presentation
img/misc/local-community.png	Empty alt, not marked presentation
img/misc/employment-carreers.png	Empty alt, not marked presentation
img/misc/visit-hike.png	Empty alt, not marked presentation

B. Heading Hierarchy

Level	Text	Status
H1	(missing)	FAIL
H3	Text Size / Contrast / Appearance / Chatbot	WARN Starts at H3
H3	[Hero image caption - rotates]	OK
H3	I am:	OK
H2	Featured Services	OK
H3	Vehicle Renewal / Hunting / Driver License / Jobs / Vital Records / View More	OK
H2	Utah Local	OK
H3	My Representatives / Meetings / Jobs	OK
H2	Visiting Utah	OK
H3	Travel & Tourism / Utah State Parks / View More	OK

C. Touch Target Violations (22 of 102 elements)

Element	Size (px)	Required
"Here is how you know" button	139 × 16	44 × 44
Logo link	229 × 43	44 × 44
"UtahID Sign In" button	163 × 30	44 × 44
Hamburger menu button	36 × 36	44 × 44
"An outdoor person" pill	205 × 38	44 × 44
"Self Employed" pill	169 × 38	44 × 44
"Parenting Teens" pill	184 × 38	44 × 44
"Moving to Utah" pill	178 × 38	44 × 44
"Turning 18" pill	144 × 38	44 × 44
"Skip to main content" link	270 × 36	44 × 44

This report was generated using automated tools (axe-core 4.7.2) supplemented with manual accessibility testing.

For questions or to begin the migration process, contact your Adobe EDS team.